

LECTURE 14

Wednesday
18/3/26

= 0.0067 Ans.

- i, Distribution function
- ii, Expected value, variance
- iii, Standard deviation
- iv, Median and Mode of the probability distribution function.

⇒ **Distribution Function:** (c.d.f of discrete)

"The distribution function of a random variable X , denoted by $F(x)$, is defined by $F(x) = P(X \leq x)$. The function $F(x)$ gives the probability of the event that X takes a value less than or equal to a specified value. for it to be CDF, there must be equal sign with the equality

It is also called **cumulative distribution function (c.d.f).**"

$f(x)$ is Prob. Distribution Func. (PDF) from last lecture and here is $F(x)$ which is CDF

$$P(X \leq 10) = P(X=0) + P(X=1) + P(X=2) + \dots + P(X=10)$$

Theorem: It is a CDF if it satisfies these 3 conditions

- $F(-\infty) = 0$
- $F(+\infty) = 1$
- If $a < b$ then $F(a) \leq F(b)$ for any real number 'a' and 'b'.

Q: The probability distribution

$$f(x) = \frac{1}{16} \binom{4}{x}, \quad x = 0, 1, 2, 3, 4. \quad \text{Find the}$$

cumulative distribution function of the r.v. X and also draw its graph.

Sol: $f(x) = P(X=x)$

$$f(0) = P(X=0) = \frac{1}{16} {}^4C_0 = \frac{1}{16}$$

$$f(1) = P(X=1) = \frac{1}{16} {}^4C_1 = \frac{4}{16}$$

$$f(2) = P(X=2) = \frac{1}{16} {}^4C_2 = \frac{6}{16}$$

$$f(3) = P(X=3) = \frac{1}{16} {}^4C_3 = \frac{4}{16}$$

$$f(4) = P(X=4) = \frac{1}{16} {}^4C_4 = \frac{1}{16}$$

finding discrete values for 'equal to' part of CDF using PDF

$$F(x) = P(X \leq x) = P(X < x) + P(X = x)$$

$$F(0) = P(X \leq 0) = \frac{1}{16}$$

by formula of CDF $F(0) = F(X=0) + F(X < 0)$

$$F(0) = \frac{1}{16} + 0$$

$$\Rightarrow F(0) = \frac{1}{16}$$

$$F(1) = P(X \leq 1) = \frac{1}{16} + \frac{4}{16} = \frac{5}{16}$$

$$F(2) = P(X \leq 2) = \frac{5}{16} + \frac{6}{16} = \frac{11}{16}$$

$$F(3) = P(X \leq 3) = \frac{11}{16} + \frac{4}{16} = \frac{15}{16}$$

$$F(4) = P(X \leq 4) = \frac{15}{16} + \frac{1}{16} = 1$$

answer by
compiling above
calculated info

$F(x)$	0	$x < 0$	$P(X < 0) = 0$
	$1/16$	$0 \leq x < 1$	$P(X \leq -1.5) = 0$
	$5/16$	$1 \leq x < 2$	$P(X < -2) = 0$
	$11/16$	$2 \leq x < 3$	$P(X \leq 5) = 1$
	$15/16$	$3 \leq x < 4$	$P(X \leq 5) + P(X = 5)$
	1	$x \geq 4$	$P(X \leq 10) = 1$

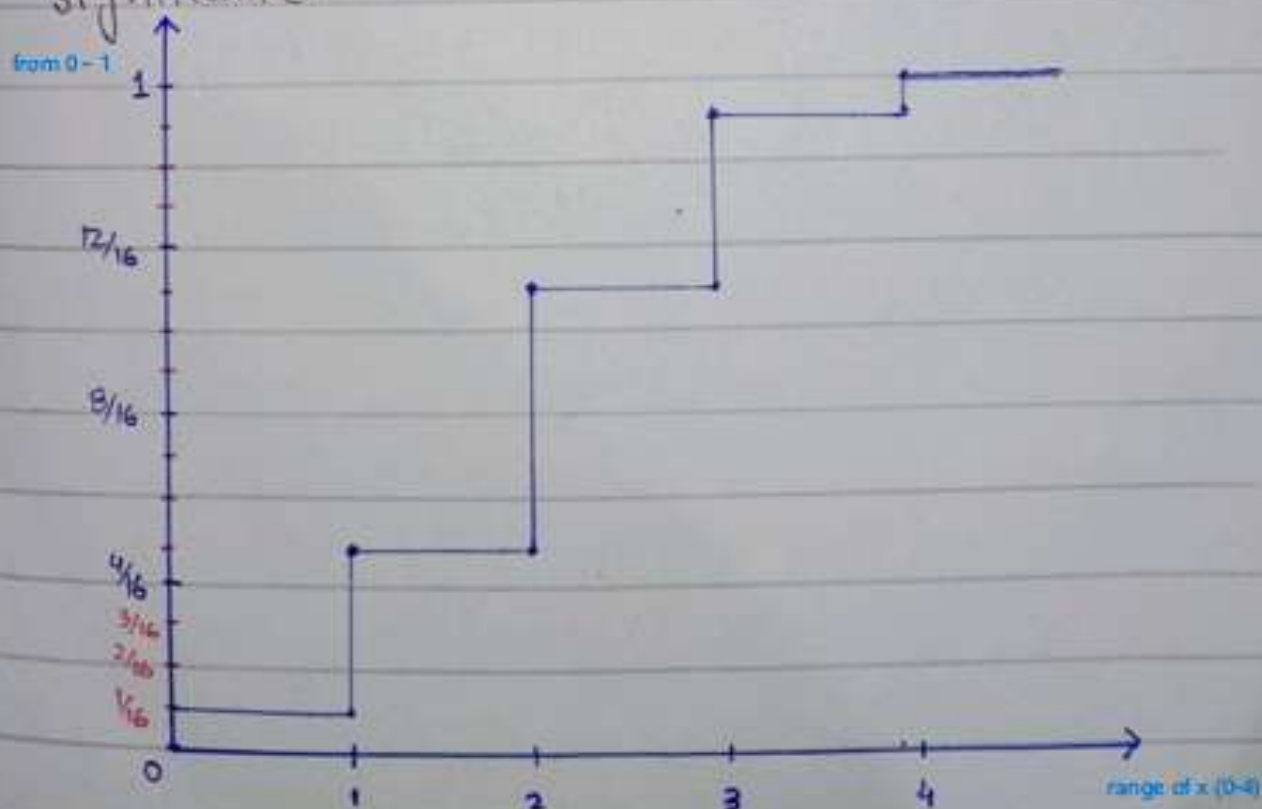
By Definition

Domain $F(x) = -\infty < x < +\infty$

Range $F(x) \in [0, 1]$

Note:

The cdf is defined not only for the values taken on by the r.v but for all real numbers. Although the probability of getting $F(1.3)$ may not be of any real significance.



Now we can find $P(X \leq 2) = P(X < 2) + P(X = 2)$

But $P(X \geq 3) = ?$

To find this we convert it into

$$F(x) = P(X \leq x)$$

and for this we write it as

$$P(X \geq x) = 1 - P(X < x) \quad \text{not a CDF since no "equal to"}$$

o) Finding the probability using c.d.f for discrete r.v X. **-> RULES**

i, $P(a < X \leq b) = F(b) - F(a)$

iv, $P(a \leq X \leq b) = F(b) - F(a-1)$

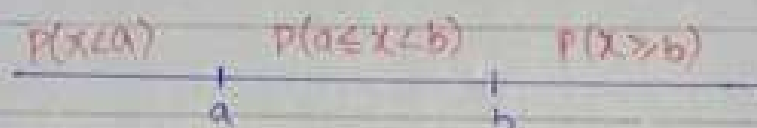
iii, $P(a \leq X < b) = F(b-1) - F(a-1)$

iv, $P(a < X < b) = F(b-1) - F(a)$

v, $P(X = a) = F(a) - F(a-1)$

optional proof

iii, Sol:



$$P(X < a) + P(a \leq X < b) + P(X > b) = 1$$

$$P(a \leq X < b) = 1 - P(X > b) - P(X < a)$$

$$= P(X < b) - P(X < a)$$

$$= P(X \leq (b-1)) - P(X \leq (a-1))$$

$$= F(b-1) - F(a-1)$$

Q. Given the discrete r.v. X has the cdf

$$F(x) = \begin{cases} 0 & x < 1 \\ 1/3 & 1 \leq x < 4 \\ 1/2 & 4 \leq x < 6 \\ 5/6 & 6 \leq x < 10 \\ 1 & x \geq 10 \end{cases}$$

Find the probability: using rules on prev. page

i, $P(2 < X \leq 6)$ (v) $P(3 \leq X < 11)$

ii, $P(X=4)$

iii, $P(2 \leq X < 6)$

iv, $P(2 \leq X \leq 6)$

Sol: (i) $P(a < X \leq b) = F(b) - F(a)$

$$= F(6) - F(2)$$

$$= \frac{5}{6} - \frac{1}{3} = \frac{3}{6} = \frac{1}{2}$$

(ii) $P(X=4)$

$$P(X=4) = F(a) - F(a-1)$$

$$= F(4) - F(4-1)$$

$$= F(4) - F(3)$$

$$= \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

(iii) $P(2 \leq X < 6)$

$$P(2 \leq X < 6) = F(b-1) - F(a-1)$$

$$= F(6-1) - F(2-1)$$

$$= F(5) - F(1)$$

$$= \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

$$\begin{aligned} \text{(iv)} \quad P(2 \leq X \leq 6) &= F(b) - F(a-1) \\ &= F(6) - F(2-1) \\ &= F(6) - F(1) \\ &= \frac{5}{6} - \frac{1}{3} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{(v)} \quad P(3 \leq X < 11) &= F(b-1) - F(a-1) \\ &= F(10) - F(2) \\ &= 1 - \frac{1}{3} \\ &= \frac{2}{3} \end{aligned}$$

C.d.f for Continuous r.v :

Definition:

limit will be min to x

If X is a continuous r.v the function given by $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$

Prob Density Func. to CDF

where $-\infty < x < +\infty$

where the $f(t)$ is the value of prob. density function of X at t

same
form
for all
cases

Probability formula using CDF for Continuous R.V

$$P(a \leq x \leq b) = F(b) - F(a)$$

$$f(x) = \frac{dF(x)}{dx}$$

CDF to Prob Density Func.

Note:

If a R.V is continuous then

$$P(a \leq x < b) = P(a \leq x \leq b) = P(a < x \leq b) = P(a < x < b)$$

⇒ For continuous:

$$P(x \leq a) = P(x < a) + P(x = a)$$

$$P(x \leq a) = P(x < a) + 0$$

$$\text{i.e. } P(x = 60) = 0$$

$$P(X \leq 7) = P(X < 7) + P(X = 7)$$

$$= P(X < 7) + 0$$

$$P(X \leq 7) = P(X < 7)$$

⇒ For Discrete:

$$P(X \leq 1) = P(X < 1) + P(X = 1)$$

o) Finding the probability using c.d.f for continuous r.v X.

i, $P(X \leq a) = F(a)$

ii, $P(X \geq a) = 1 - F(a)$

iii, $P(a \leq X \leq b) = F(b) - F(a)$

iv, $P(X \leq a) = P(X < a)$

(From pdf to cdf)

Q: For the given probability density function $f(x) = 3e^{-3x}$, $0 < x < \infty$

i, Find c.d.f

ii, Using c.d.f, evaluate the following probabilities.

a) $P(X < 1.5)$

(b) $P(X \geq 0.5)$

c) $P(0.5 < X < 1.)$

Sol: (i) $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$ By DEF. formula

$$= \int_0^x 3e^{-3t} dt = 3 \left| \frac{e^{-3t}}{-3} \right|_0^x$$

$$= \frac{3}{3} \left| e^{-3t} \right|_0^x$$

$$= (e^0 - e^{-3x})$$

$$F(x) = 1 - e^{-3x}$$

OR

in cdf limit
will be min to
x.

$$0 < x < \infty$$

$$F(0) = 0$$

$$F(\infty) = 1$$

$$-\infty < x < \infty$$

$$F(-\infty) = 0$$

$$F(\infty) = 1$$

$$= 3 \left| \frac{e^{-3t}}{-3} \right|_0^x = 3 \left[\frac{e^{-3(x)}}{-3} - \frac{e^{-3(0)}}{-3} \right]$$

$$= 3 \left[\frac{e^{-3x}}{-3} - \frac{e^0}{-3} \right] = 3 \left[\frac{e^{-3x}}{-3} - \frac{1}{-3} \right]$$

$$= 3 \left[\frac{e^{-3x}}{-3} + \frac{1}{3} \right] = 3 \left[\frac{1}{3} - \frac{e^{-3x}}{3} \right]$$

$$= \frac{3}{3} [1 - e^{-3x}] = [1 - e^{-3x}]$$

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-3x} & x \geq 0 \end{cases}$$

(ii) (a) $P(x < 1.5)$

$$P(x < 1.5) = F(a)$$

$$P(x < 1.5) = 1 - e^{-3x}$$

$$= 1 - e^{-3(1.5)}$$

$$P(x < 1.5) = 0.989$$

(b) $P(x \geq 0.5) = 1 - F(a)$

$$= 1 - (1 - e^{-3x})$$

$$= 1 - 1 + e^{-3x}$$

$$= e^{-3(0.5)}$$

$$P(x \geq 0.5) = 0.2231$$

(c) $P(0.5 < x < 1) = F(b) - F(a)$

$$= (1 - e^{-3b}) - (1 - e^{-3a})$$

$$= [1 - e^{-3(1)}] - [1 - e^{-3(0.5)}]$$

$$= (1 - e^{-3}) - (1 - e^{-1.5})$$

Converted first since it was "greater" instead of "less than"

$$= (1 - e^3 - 1 + e^{-1.5})$$

$$= 0.173$$

→ From cdf to pdf:

$$f(x) = \frac{d(F(x))}{dx}$$

$$cdf = \begin{cases} 0 & x < 0 \\ 1 - e^{-3x} & x \geq 0 \end{cases}$$

$$F(x) = 1 - e^{-3x}$$

$$d(cdf) = 0 - (-3)e^{-3x}$$

$$(pdf) \Rightarrow f(x) = 3e^{-3x}$$

-----Rest Practice ques. in GCR Notes -----

Expectation of random variable: $(E(X))$

$$E(X) = \text{Average} = \text{Mean}$$

Discrete

$$E(X) = \sum x f(x)$$

$$E(g(x)) = \sum g(x) f(x) \quad (\text{for any given function})$$

e.g: $E(2x) = \text{SUM}(2x)(fx)$

Continuous

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$E(g(x)) = \int_{-\infty}^{\infty} g(x) \cdot f(x) dx$$

(for any given function)

7.22

Q: (a) The p.d. of a discrete random variable X as

put $x=0$ $f(x) = \binom{3}{x} \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{3-x}, x=0,1,2,3$

Find $E(X), E(X^2)$. this is $g(x)$ which could be any func.

Sol:

x	$f(x)$	$x f(x)$	$x^2 f(x)$
0	0.421875	0	0
1	0.421875	0.42185	0.42185
2	0.140625	0.28125	0.5625
3	0.015625	0.046875	0.140625
Σ	1	0.75	1.125

Hence $E(X) = \sum x f(x) = 0.75$ and

$$\Sigma(X^2) = \sum x^2 f(x) = 1.125 \quad E(g(x)) = \text{SUM}(g(x))(fx)$$

Continuous R.V

Q: let X be r.v with p.d.f

$$f(x) = \begin{cases} 2(x-1) & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find expected value of $g(x)$ where

i) $g(x) = x$

(ii) $g(x) = x^2$

(iii) $g(x) = 2x - 1$

Sol: (i) $E(g(x)) = \int_{-\infty}^{\infty} g(x)f(x)dx$

$$E(g(x)=x) = \int_1^2 x \cdot f(x) dx$$

$$\begin{aligned} E(x) &= \int_1^2 x \cdot 2(x-1) dx \\ &= 2 \int_1^2 (x^2 - x) dx \end{aligned}$$

$$= 2 \left[\int_1^2 x^2 dx - \int_1^2 x dx \right]$$

$$= 2 \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_1^2$$

$$= 2 \left[\frac{(2)^3}{3} - \frac{(2)^2}{2} - \left(\frac{(1)^3}{3} - \frac{(1)^2}{2} \right) \right]$$

$$= 2 \left[\frac{8}{3} - 2 - \frac{1}{3} + \frac{1}{2} \right] = \frac{5}{6}$$

(ii) $E(g(x)) = \int_{-\infty}^{\infty} g(x)f(x)dx$

$$E(g(x)=x^2) = \int_1^2 x^2 \cdot 2(x-1) dx$$

$$= 2 \int_1^2 (x^3 - x^2) dx$$

$$= 2 \left[\frac{x^4}{4} - \frac{x^3}{3} \right]_1^2$$

$$= 2 \left[\frac{(2)^4}{4} - \frac{(2)^3}{3} - \frac{(1)^4}{4} + \frac{(1)^3}{3} \right]$$

$$2 \left[4 - \frac{8}{3} - \frac{1}{4} + \frac{1}{3} \right] = \frac{17}{6}$$

$$(iii) E(2x-1) = 2 \int_1^2 (2x-1)(x-1) dx$$

$$= 2 \int_1^2 (2x^2 - 2x - x + 1) dx$$

$$= 2 \left[\frac{2x^3}{3} - \frac{2x^2}{2} - \frac{x^2}{2} + x \right]_1^2$$

$$= 2 \left[\frac{2x^3}{3} - \frac{3x^2}{2} + x \right]_1^2$$

$$= 2 \left[\frac{2}{3} (2)^3 - \frac{3}{2} (2)^2 + 2 - \frac{2}{3} + \frac{3}{2} - 1 \right]$$

$$= 2 \left[\frac{16}{3} - 12 + 2 - \frac{2}{3} + \frac{3}{2} - 1 \right] = \frac{7}{3}$$

Formulas:

• μ Mean = $E(x)$

• σ^2 Variance = $\text{Var}(x) = E(x^2) - (E(x))^2$

• σ Standard deviation = S.D.(x) = $\sqrt{E(x^2) - (E(x))^2}$

$$\text{S.D.}(x) = \sqrt{\text{Var}(x)}$$

Q: If continuous r.v of X has p.d.f

$$f(x) = \begin{cases} \frac{3}{4}(3-x)(x-5) & 3 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

Find the arithmetic Mean, variance and standard variation.

Sol: $E(X) = \frac{3}{4} \int_3^5 x(3-x)(x-5) dx$ $E(X) = \int_{-\infty}^{\infty} x f(x) dx$

$$= \frac{3}{4} \int_3^5 x(3x-15-x^2+5x) dx$$

$$= \frac{3}{4} \int_3^5 x(-x^2+8x-15) dx$$

$$= \frac{3}{4} \int_3^5 (-x^3+8x^2-15x) dx$$

$$= \frac{3}{4} \left[\frac{-x^4}{4} + \frac{8x^3}{3} - \frac{15x^2}{2} \right]_3^5$$

$$= \frac{3}{4} \left[\frac{-(5)^4}{4} + \frac{8(5)^3}{3} - \frac{15(5)^2}{2} + \frac{(3)^4}{4} - \frac{8(3)^3}{3} + \frac{15(3)^2}{2} \right]$$

$$= 4$$

Again $E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$

$$= \frac{3}{4} \int_3^5 x^2(3-x)(x-5) dx$$

$$= \frac{3}{4} \int_3^5 (-x^4+8x^3-15x^2) dx$$

$$= \frac{3}{4} \left[\frac{-x^5}{5} + \frac{8x^4}{4} - \frac{15x^3}{3} \right]_3^5$$

$$= \frac{3}{4} \left[\frac{-1}{5} (3125) + 2(625) - 5(125) - \left\{ \frac{-1}{5} (243) + 2(81) - 5(27) \right\} \right]$$

$$= \frac{3}{4} \left[0 + \frac{243}{5} - 162 + 135 \right]$$

$$= \frac{3}{4} \left[\frac{108}{5} \right] = \frac{81}{5}$$

$$\therefore \text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \frac{81}{5} - (4)^2 = \frac{1}{5} = 0.2, \text{ Mean} = 4$$

$$\text{S.D.}(X) = \sqrt{\frac{1}{5}} = \sqrt{0.2} = 0.447$$

7.22

Q. 16) Let X be a random variable with probability distribution.

x	-1	0	1	2	3
$f(x)$	0.125	0.50	0.20	0.05	0.125

i) Find $E(X)$ and $\text{Var}(X)$.

Sol:

x	$f(x)$	$xf(x)$	$x^2f(x)$
-1	0.125	-0.125	0.125
0	0.500	0.000	0.000
1	0.200	0.200	0.200
2	0.050	0.100	0.200
3	0.125	0.375	1.125
Total	1.000	0.550	1.650

Thus $E(x) = \sum x f(x) = 0.55$
 and $\text{Var}(X) = E(X^2) - [E(X)]^2$

$$= 1.650 - (0.55)^2$$

$$= 1.650 - 0.3025$$

$$= 1.3475$$

put values of x from
above table into eq.

(ii) Find the p.d. of the n.v $Y = 2X+1$. Using the p.d. of Y, determine $E(Y)$ and $\text{Var}(Y)$.

Sol: Computation of the p.d. of $Y = 2X+1$. $E(Y)$ and $\text{Var}(Y)$.

same as $f(x)$ since prob. does not changes

$y = (2x+1)$	$f(y)$	$y f(y)$	$y^2 f(y)$
-1	0.125	-0.125	0.125
1	0.500	0.500	0.500
3	0.200	0.600	1.800
5	0.050	0.250	1.250
7	0.125	0.875	6.125
Total	1.000	2.100	9.800

$$\therefore E(Y) = \sum y f(y) = 2.1 \text{ and}$$

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = 9.80 - (2.1)^2$$

$$= 9.80 - 4.41 = 5.39$$

Mode is the x value in front of the maximum value of $f(x)$

Formulas:

Median = $F(x) = \frac{1}{2}$

equate CDF to $\frac{1}{2}$
(for continuous)

Mode = Most frequent value

(point of maxima in function)

Q: Find Median and Mode for the following probability distribution.

$$f(x) = \binom{3}{x} \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{3-x}$$

$x = 0, 1, 2, 3$

Discrete R.V
(median, mode)

Sol:

x	$f(x)$	$F(x)$ (CDF)
mod 0	max $f(x)$ 0.4218	0.4218
mod 1	max $f(x)$ 0.4218	$0.4218 + 0.4218 = 0.8436$
2	0.1406	$0.8436 + 0.1406 = 0.984$
3	0.0156	$0.984 + 0.0156 = 0.9996$
Σ	1	

x	$F(x)$	$f(x)$	Median group for value of x
$F(1)$	1	0.8436	0.4218

So,

✓ Median = 1

FOR DISCRETE MEDIAN since in front of $x=1$, we have $F(x) = 0.8436$ which was covering $\frac{1}{2}$ (by def. of median) which wasn't covered by $x = 0$ since it has $F(x) = 0.4218$

$\Rightarrow$ Mode:

If X is a discrete r.v then the mode is value of x of maximum probability of X is a continuous r.v then we get a maximum value of $f(x)$ as

$$(i) \frac{d}{dx} f(x) = 0, \quad (ii) f''(x) < 0$$

Provided that the solution of $f'(x) = 0$ lies within the given range of the r.v. X .

Continuous R.V (median, mode, SD)

Q: Let X be r.v with p.d.f.

$$f(x) = K(x - x^2), \quad 0 \leq x \leq 1$$

where K is constant then find median, mode, mean and standard deviation.

Sol: First we will find value of K .

$$\int_0^1 f(x) dx = K \int_0^1 (x - x^2) dx$$

$$1 = K \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1$$

$$1 = K \left[\frac{1}{2} - \frac{1}{3} \right]$$

$$1 = K \left(\frac{1}{6} \right) \Rightarrow K = 6$$

$$f(x) = 6(x - x^2) \quad 0 \leq x \leq 1$$

$$F(x) = 6 \int_0^x (x - x^2) dx$$

$$= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^x$$

$$= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right] = 6 \left(\frac{3x^2 - 2x^3}{6} \right)$$

$$F(x) = 3x^2 - 2x^3 \quad \text{--- (i)}$$

Put $F(x) = \frac{1}{2}$ in (i)

$$\frac{1}{2} = 3x^2 - 2x^3$$

$$3x^2 - 2x^3 - \frac{1}{2} = 0$$

$$2x^3 - 3x^2 + \frac{1}{2} = 0$$

0.5 is the root of above eq

0.5	4	-6	0	1
	↓	2	-2	-1
	4	-4	-2	0

$$4x^2 - 4x - 2 = 0$$

$$a = 4, \quad b = -4, \quad c = -2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-4) \pm \sqrt{16(16)(-2)}}{2(4)}$$

$$= \frac{4 \pm \sqrt{16+32}}{8} = \frac{4 \pm \sqrt{48}}{8}$$

$$\Rightarrow \frac{1+\sqrt{3}}{2}, \quad \frac{1-\sqrt{3}}{2}$$

$$\Rightarrow 1.366025, \quad -0.366$$

But both values -0.366 and 1.366 are unacceptable since 'x' must lie in the interval $(0,1)$. The median is therefore given by $x = \frac{1}{2}$.

The mode is the value of x for which

(i) $f'(x) = 0$ and (ii) $f''(x) < 0$

$$f(x) = 6(x-x^2), \quad \text{and} \quad f'(x) = 6(1-2x)$$

Now $f'(x) = 0$ when $1-2x = 0$ or $x = \frac{1}{2}$